

Azure Synapse Analytics – One-Page Cheat Sheet

1. Components

- Dedicated SQL Pool (MPP Data Warehouse)
- Serverless SQL Pool (Pay-per-query)
- Apache Spark Pool (Big Data + ML)
- Data Explorer Pool (Kusto engine for telemetry)
- Synapse Pipelines (ADF-based ETL/ELT)
- Synapse Studio (Unified interface)
- ADLS Gen2 Integration (Workspace storage)

2. Dedicated SQL Pool

- Distribution: Hash, RoundRobin, Replicate
- Indexes: Clustered, Clustered Columnstore, Nonclustered
- Performance: Resource classes, Materialized views, CTAS
- Pipeline loading via PolyBase or COPY INTO

3. Serverless SQL Pool

- Query ADLS directly (CSV/JSON/Parquet)
- OPENROWSET for ad-hoc queries
- External tables supported
- Best for BI exploration, lightweight transformations

4. Apache Spark Pool

- Languages: PySpark, Scala, SparkSQL, .NET
- Use Delta/Parquet for performance
- Supports MLlib, structured streaming
- Avoid small files, use OPTIMIZE/MERGE

5. Synapse Pipelines

- Activities: Copy, Notebook, SQL Stored Proc, REST, Mapping Data Flows
- IR Types: AutoResolve, Azure, Self-hosted
- Supports triggers (schedule, event, tumbling window)

6. Security

- AD Authentication, RBAC, SQL Permissions
- Private Endpoints + Managed VNET
- TDE for SQL, SSE for Storage
- Managed Identity for pipelines & notebooks

7. Lake Database

- Delta Lake tables stored in ADLS
- Queried by both Spark + Serverless SQL
- Schema governed & versioned

8. Cost Optimization

- Pause Dedicated SQL Pool when idle
- Prefer Parquet/Delta over CSV
- Minimize serverless scans (use predicates)
- Use small resource classes for non-prod

9. Best Use Cases

- Dedicated SQL → Enterprise DWH
- Serverless SQL → Quick ad-hoc analytics
- Spark → Data engineering + ML
- Data Explorer → Telemetry & real-time logs
- Pipelines → Orchestration & ELT

10. Interview Keywords

- MPP, Columnstore, Distribution Key, PolyBase, Delta Lake
- Pipelines vs ADF, Serverless vs Dedicated SQL
- Lakehouse architecture, RBAC, Private Link